

Noel Joyce Varghese

noeljoyce6@gmail.com

+91 8590607872

linkedin.com/in/noel-joyce-varghese

github.com/noeljoyce6

Professional Summary

Embedded Hardware Design Engineer specializing in embedded electronic systems, PCB design, mixed-signal electronics, power electronics, and hardware validation for robotics, aerospace, defence, and autonomous platforms. Experienced in schematic design, multilayer PCB layout, board bring-up, system integration, signal/power integrity considerations, and production-ready hardware development.

Education

B.Tech in Electronics and Communication Engineering

Amal Jyothi College of Engineering (Autonomous)

Oct 2022 – Apr 2026

Kerala, India

Professional Experience

PCB Design Intern

PSYC Aerospace and Defence Industries Pvt. Ltd.

28th Nov 2025 – Present

Bengaluru, India

- Engineered production-grade hardware for aerospace and robotics applications across 8+ embedded hardware projects including schematic capture, PCB layout, and BOM generation.
- Integrated sensors and communication interfaces (SPI/I2C/UART/CAN) into multiple hardware platforms for embedded control and telemetry.
- Executed board bring up, debugging, validation, soldering/rework, and fabrication release support.

Analog Design Intern

IIIT Kottayam

Jun 2025

Kerala, India

- Simulated a 5 GHz current-starved CMOS VCO using LTspice and Cadence Virtuoso.
- Analyzed analog circuits and transistor-level RF design fundamentals for oscillator and high-frequency design applications.

Skills

- Hardware Design:** PCB Layout, Multi-layer, Mixed-Signal, High-Current Routing, DFM/DFA
- PCB Design:** Controlled Impedance, BOM Generation, Gerber Release
- EDA Tools:** Altium, KiCad, OrCAD, Virtuoso, LTspice
- Embedded Platforms:** STM32, ESP32, Raspberry Pi, Arduino
- Interfaces:** SPI, I2C, UART, CAN, USB, ADC/DAC
- Power Electronics:** DC-DC, LDOs, Motor Drivers, Protection Circuits
- Programming:** C, C++, Python
- Dev Tools:** Git, VS Code, Arduino IDE
- Validation:** Bring-Up, Oscilloscope, Logic Analyzer, Debugging
- Standards:** IPC-2221, IPC-7351, EMC/EMI Awareness

Technical Projects

Unified Field-Oriented Control (FOC) Motor Controller Board

- Architected a 4-layer 50mm × 50mm BLDC motor controller PCB using STM32G431 and DRV8353R for real-time FOC applications.
- Implemented current sensing, protection circuitry, braking, and CAN-based closed-loop control interfaces.

Minima Flight Controller using CM5 (Ongoing)

- Architecting a compact 6-layer flight controller PCB around CM5/Raspberry Pi compute platform for UAV applications.
- Implementing high-density routing, power distribution, sensor integration, and peripheral interfacing in constrained form factor.

Nemo Winch V3.0 – Drone Payload Delivery Control Board

- Created control PCB for UAV payload winch with motor drive, encoder feedback, servo actuation, and Pixhawk interfacing.

Automated Security and Deterrent System – Detection/Actuation Board

- Integrated ESP32-CAM, mmWave radar, PIR sensing, servo actuation, and dual-rail power gating into surveillance/actuation PCB.

Wireless 4-Channel EEG Acquisition System Board

- Built low-noise mixed-signal EEG acquisition hardware with AD623 analog front-end and BLE wireless streaming.

Smart Helmet for Accident & Health Monitoring

- Integrated ESP32, MAX30105, MPU6050, GPS, GSM, and IR sensors into wearable accident/health monitoring platform.
- Performed EEG-based health simulation and signal analysis in MATLAB.

Vanguard MK1 – Search & Rescue Crawling Robot (Academic Project)

- Designed control circuitry for a 12-DOF crawling robot using Raspberry Pi 5, PCA9685, MG90 servos, and ToF LiDAR.
- Developing control PCB for locomotion, sensor interfacing, and power distribution.

Leadership / Technical Activities

- Led a hands-on PCB workshop for ECE students involving schematic design, KiCad PCB layout, fabrication, and soldered assembly of a mini audio amplifier board in collaboration with the ECE department.
- Core Team Member for NASA Space Apps Challenge, supporting technical planning and execution across 2 annual editions.
- Active volunteer under IEDC / Kerala Startup Mission, contributing to innovation-driven technical initiatives and hardware-focused workshops.